Motion Physio Clinic

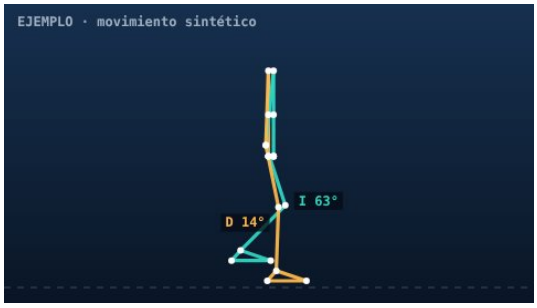
Video-based biomechanical analysis report
10 October 2026

Ana López

31 years · female · 166 cm · ACL rehabilitation, left knee

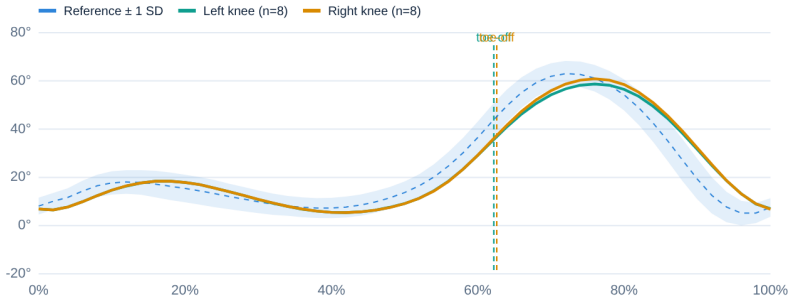
Gait · Session 5 of 5 · Side view
Capture quality 97 % · 16 strides

WALKING SPEED	STATUS	VS. PREVIOUS SESSION	VS. FIRST SESSION
1.28 ± 0.05 m/s	**No alerts**	**+0.07 m/s (p 0.001)**	**+0.37 m/s (p < 0.001)**



Frame of maximum knee flexion

Knee flexion over the gait cycle (degrees)



Mean of all cycles per side against the laboratory reference of Bovi et al. (2011): adults 22–72 years, slow speed. Vertical lines: mean toe-off.

Cycle phases (left stride)



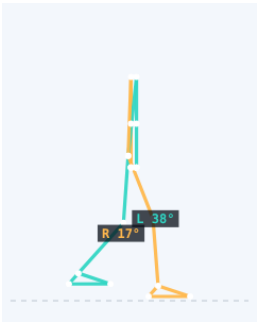
Initial contact
0 % of cycle



Loading response
12 % of cycle



Mid-stance
30 % of cycle



Toe-off
62 % of cycle



Mid-swing
85 % of cycle

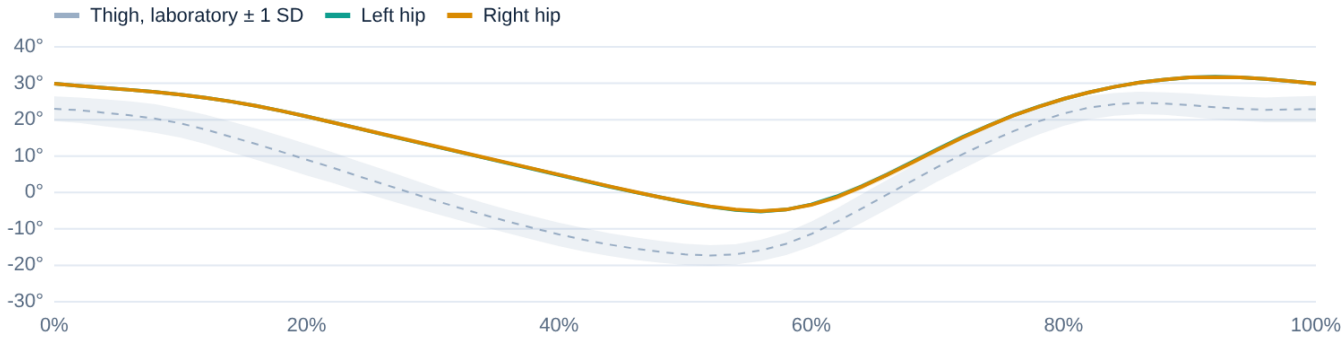
Results

Parameter	Mean ± SD	n	Reference	Assessment	Rel.
SPATIOTEMPORAL					
Walking speed	**1.28 ± 0.05 m/s**	16	>= 1.14 m/s (mean 1.34 for age and sex - 15 %)	In range	High
Cadence	**112 ± 2 steps/min**	16	96–123 (laboratory)	In range	High
Stride length	**1.36 ± 0.05 m**	16	0.72–0.92 × height	In range	High
Stride time	**1.07 ± 0.02 s**	16	0.95–1.25 s	In range	High
Left stance phase	**62.2 ± 0.4%**	8	60–67 % (laboratory)	In range	Medium
Right stance phase	**62.6 ± 0.6%**	8	60–67 % (laboratory)	In range	Medium
Double support	**24.7 ± 0.5%**	16	20–32 % (laboratory)	In range	Medium
SYMMETRY AND VARIABILITY					
Stride variability (CV)	**1.6%**	16	< 3 %	In range	High
Step time asymmetry (SI)	**0.1%**	8	< 10 %	In range	High
Stance phase asymmetry (SI)	**0.5%**	8	< 10 %	In range	High
Step length asymmetry (SI)	**0.1%**	8	< 10 %	In range	High
Swing knee flexion asymmetry (SI)	**3.6%**	8	< 10 %	In range	High
SAGITTAL KINEMATICS					
Left knee: peak swing flexion	**59 ± 3°**	8	52–74° (laboratory)	In range	High

Parameter	Mean ± SD	n	Reference	Assessment	Rel.
Left knee: loading response flexion	**18 ± 0°**	8	8–28° (laboratory)	**In range**	High
Left knee: flexion at initial contact	**7 ± 0°**	8	1–15° (laboratory)	**In range**	High
Left knee: deviation from the normal curve (GVS)	**6°**	8	< 8° (indicative)	**In range**	High
Left hip: sagittal range	**37 ± 0°**	8	35–49° (laboratory), indicative	**In range**	High
Right knee: peak swing flexion	**61 ± 2°**	8	52–74° (laboratory)	**In range**	High
Right knee: loading response flexion	**18 ± 0°**	8	8–28° (laboratory)	**In range**	High
Right knee: flexion at initial contact	**7 ± 0°**	8	1–15° (laboratory)	**In range**	High
Right knee: deviation from the normal curve (GVS)	**6°**	8	< 8° (indicative)	**In range**	High
Right hip: sagittal range	**37 ± 0°**	8	35–49° (laboratory), indicative	**In range**	High
Trunk lean (mean)	**4°**				High

Reliability: high = suitable view, good visibility and enough cycles or repetitions; medium = few cycles, oblique view or a variable sensitive to 2D error; low = indicative only, raises no alerts.

Hip flexion over the cycle (± 1 SD between cycles)

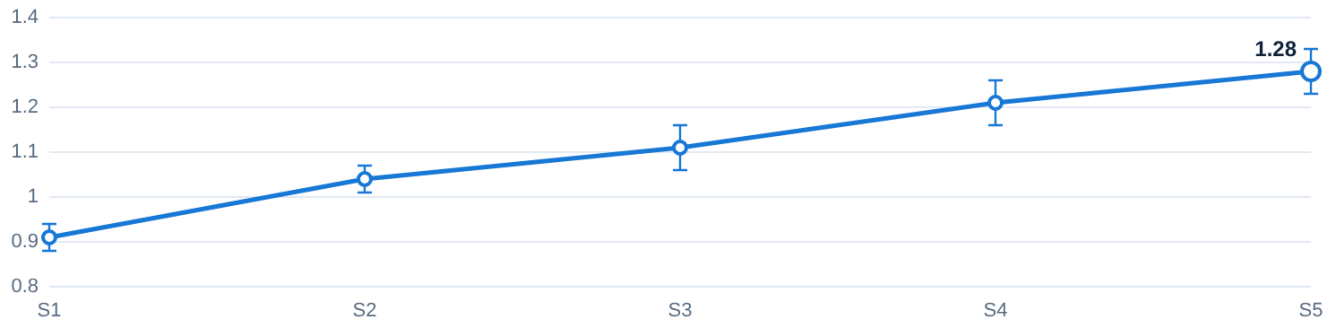


Comparison with the previous session (30 September 2026)

Parámetro	Anterior	Actual	Dif.	p	d	
Walking speed	1.21 m/s	1.28 m/s	+0.06	0.001	1,3	No relevant change
Cadence	108.49 steps/min	112.47 steps/min	+3.97	< 0.001	1,9	No relevant change
Stride time	1.11 s	1.07 s	-0.04	< 0.001	-1,9	No relevant change
Stride length	1.34 m	1.36 m	+0.02	0.288	0,4	No relevant change
Left stance phase	61.79%	62.24%	+0.45	0.053	1,1	No relevant change
Right stance phase	62.38%	62.57%	+0.2	0.548	0,3	No relevant change
Double support	24.2%	24.68%	+0.48	0.010	1,0	No relevant change
Left knee: peak swing flexion	55.32°	58.72°	+3.39	0.073	1,1	No relevant change
Right knee: peak swing flexion	60.92°	60.88°	-0.04	0.972	-0,0	No relevant change
Left knee: loading response flexion	18.43°	18.46°	+0.03	0.172	0,7	No relevant change
Right knee: loading response flexion	18.42°	18.45°	+0.03	0.293	0,6	No relevant change
Left hip: sagittal range	36.89°	36.92°	+0.04	0.122	0,9	No relevant change
Right hip: sagittal range	36.89°	36.89°	-0.01	0.901	-0,1	No relevant change

p: Welch t-test between the cycles or repetitions of both sessions. d: Cohen's effect size. Improvement or deterioration is only reported if p < 0.05 and the difference exceeds the minimal relevant change of the variable.

Progress: walking speed by session (mean ± SD)



Linear trend: +0.06 m/s per week (p < 0.001; R2 0.99; 5 sessions).

Professional assessment

16 strides were analysed (8 left, 8 right). Walking speed 1.28 m/s ± 0.05, cadence 112 steps/min ± 2. Peak knee flexion in swing: 59° ± 3 (left) and 61° ± 2 (right). The parameters with an available reference are within range. Values obtained by 2D computer vision; they should be interpreted according to the physiotherapist's judgement.

Method and measurement quality

Side view (index 0.07) · person detected in 100 % of 300 frames · mean visibility 90 % · 30 Hz · duration 10 s · 2 left/right corrections.

• 2D pose estimation with MediaPipe BlazePose (2 landmarks; Bazarevsky et al., 33), synthetic model.
• Uniform resampling at 30 Hz; landmarks with visibility < 0.5 discarded and gaps <= 0.25 s linearly interpolated.
• Left/right swap correction by temporal continuity (2 frames corrected).
• 2nd-order low-pass Butterworth filter, forward and backward (zero phase, effective 4th order), cut-off frequency 6 Hz with Winter's correction (2009).
• Each metric is expressed as mean ± standard deviation across cycles, repetitions or windows (n). A change between sessions is considered real when it is statistically significant (Welch t-test, p < 0.05; Welch, 1947) and also exceeds the minimal relevant change of that metric (e.g. 0.10 m/s for speed; Perera et al., 2006).
• Gait events with the coordinate method of Zeni et al. (2008): initial contact = most anterior heel position relative to the pelvis; toe-off = most posterior forefoot position. Only straight walking bouts in one direction are used (turns and stops are excluded).
• Angles normalised to 0–100 % of the cycle (51 points) and averaged per side. Deviation from the reference curve is expressed as the Gait Variable Score (root mean square error; Baker et al., 2009).
• Laboratory references from Bovi et al. (2011), adults 22–72 years, slow speed, chosen from the patient's speed normalised by body height (77 % of height per second): knee and thigh curves, toe-off, double support and cadence (range = mean ± 2 SD). Kinematics depend on speed, so comparing with people walking at a similar speed avoids mistaking slow walking for an abnormality.
• Symmetry with the Robinson et al. index (1987): SI = |R - L| / (0.5·(R + L)) × 100. Variability as the coefficient of variation of stride time (Hausdorff, 2005).
• Spatial scale from body height: shoulder–ankle distance ~ 0.779 × height (Drillis and Contini, 1966). Assumes a fixed camera perpendicular to the walking direction.

References

1. Bazarevsky V, Grishchenko I, Raveendran K, et al. BlazePose: On-device real-time body pose tracking. arXiv:2006.10204; 2020.
2. Winter DA. Biomechanics and Motor Control of Human Movement. 4ª ed. Hoboken: Wiley; 2009.
3. Bovi G, Rabuffetti M, Mazzoleni P, Ferrarin M. A multiple-task gait analysis approach: kinematic, kinetic and EMG reference data for healthy young and adult subjects. Gait Posture. 2011;33(1):6-13.
4. Zeni JA, Richards JG, Higginson JS. Two simple methods for determining gait events during treadmill and overground walking using kinematic data. Gait Posture. 2008;27(4):710-4.
5. Baker R, McGinley JL, Schwartz MH, et al. The Gait Profile Score and Movement Analysis Profile. Gait Posture. 2009;30(3):265-9.
6. Robinson RO, Herzog W, Nigg BM. Use of force platform variables to quantify the effects of chiropractic manipulation on gait symmetry. J Manipulative Physiol Ther. 1987;10(4):172-6.
7. Hausdorff JM. Gait variability: methods, modeling and meaning. J Neuroeng Rehabil. 2005;2:19.
8. Perry J, Burnfield JM. Gait Analysis: Normal and Pathological Function. 2ª ed. Thorofare: SLACK; 2010.
9. Kadaba MP, Ramakrishnan HK, Wootten ME. Measurement of lower extremity kinematics during level walking. J Orthop Res. 1990;8(3):383-92.
10. Bohannon RW, Williams Andrews A. Normal walking speed: a descriptive meta-analysis. Physiotherapy. 2011;97(3):182-9.
11. Fritz S, Lusardi M. White paper: "Walking speed: the sixth vital sign". J Geriatr Phys Ther. 2009;32(2):46-9.
12. Drillis R, Contini R. Body segment parameters. Technical Report 1166-03. New York University; 1966.
13. Welch BL. The generalization of "Student's" problem when several different population variances are involved. Biometrika. 1947;34(1-2):28-35.
14. Perera S, Mody SH, Woodman RC, Studenski SA. Meaningful change and responsiveness in common physical performance measures in older adults. J Am Geriatr Soc. 2006;54(5):743-9.